EXERCISE 10

Parachutist with Linear Aerodynamic Drag and Terminal Speed

Original statement · ProblemsCh3.pdf, p. 3

Linear Drag • Terminal Velocity

PHASE 1: STATEMENT & SETUP

> A parachutist of mass m kg jumps from a plane with initial velocity 0. Because of the parachute, there is a friction force that avoids free fall; this friction turns out to be proportional (with constant of proportionality $c > 0$) to the velocity. Write down an ODE to describe the motion of the parachutist, and determine the terminal speed (when $t \to \infty$). Compare this speed with that of free fall.

PHASE 2: FUNDAMENTAL FORMULATION

Taking positive axis downwards, Newton's 2nd law gives:

$$\sum F = m\frac{dv}{dt} = mg - cv \implies \frac{dv}{dt} + \frac{c}{m}v = g$$

With initial condition $v(0) = 0$.

> METHOD OF THE CLASS NOTES [NOTES, PP. 3, 6]
>
> Velocity: $v' + \frac{c}{m}v = g, v(0) = 0$, with the three-step algorithm of the notes (p. 3): $P(t) = \int_0^t \frac{c}{m}\,d\tau = \frac{c}{m}t, I = e^{ct/m}, I(0) = 1$, and $v(t) = \frac{1}{I(t)}\int_0^t e^{c\tau/m} g\,d\tau = \frac{mg}{c}\left(1 - e^{-ct/m}\right)$.
>
> Position, as a second-order equation (p. 6): $x'' + \frac{c}{m}x' = g, x = x_H + x_N$. The roots of $k^2 + \frac{c}{m}k = 0$ are 0 and $-\frac{c}{m}$ (Case 01), so $x_H = C_1 + C_2 e^{-ct/m}$. The forcing $g = g\,e^{0t}$ has $\alpha = 0$, a root, so the educated guess of the notes is $x_N = At$: $\frac{c}{m}A = g, A = \frac{mg}{c}$, the terminal speed.

PHASE 3: STEP-BY-STEP ANALYTICAL DERIVATION

Step 1: Integrating Factor Method

Integrating factor $\mu(t) = e^{(c/m)t}$:

$$\frac{d}{dt}\left[v(t)e^{(c/m)t}\right] = ge^{(c/m)t}$$

$$v(t)e^{(c/m)t} = \int ge^{(c/m)t}\,dt = \frac{mg}{c}e^{(c/m)t} + C$$

$$v(t) = \frac{mg}{c} + Ce^{-(c/m)t}$$

Step 2: Initial Condition $v(0) = 0$

$$v(0) = \frac{mg}{c} + C = 0 \implies C = -\frac{mg}{c}$$

$$\mathbf{v(t) = \frac{mg}{c}\left(1 - e^{-(c/m)t}\right)}$$

Step 3: Terminal Velocity ($t \to \infty$)

As $t \to \infty$, $e^{-(c/m)t} \to 0$:

$$\mathbf{v_\infty = \lim_{t\to\infty} v(t) = \frac{mg}{c}}$$

Step 4: Comparison with Free Fall

- **Free fall ($c = 0$):** $v_{\text{free}}(t) = gt$. Velocity increases linearly without bound ($v \to \infty$).
- **Parachutist with Drag ($c > 0$):** Velocity saturates asymptotically to a safe finite constant $v_\infty = \frac{mg}{c}$.
- **Short-time Taylor expansion ($t \ll m/c$):**

$$v(t) = \frac{mg}{c}\left[1 - \left(1 - \frac{c}{m}t + \frac{c^2}{2m^2}t^2 - \dots\right)\right] = gt - \frac{cg}{2m}t^2 + \dots$$

At the first instant ($t \to 0$), the jumper accelerates at g, identically to free fall, before drag builds up.

PHASE 4: MASTER SUMMARY & FLIGHT MECHANICS

TERMINAL VELOCITY & VELOCITY PROFILE

$$\boxed{\frac{dv}{dt} + \frac{c}{m}v = g \implies v(t) = \frac{mg}{c}\left(1 - e^{-(c/m)t}\right)}$$

$$\boxed{v_\infty = \frac{mg}{c}}$$